

CE 776: Transportation Project Evaluation and Decision Making

Course Project, Group 30

Economic Evaluation of Delhi Metro Phase 4A



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Abstract

This report presents an economic evaluation of Delhi Metro Phase 4A, a 61.679 km, three-corridor mixed underground-and-elevated expansion of the Delhi Metro network (Aerocity–Tughlakabad, Janakpuri West–R.K. Ashram, and Mukundpur–Maujpur), including one of the network’s longest underground corridors. Using standard project-appraisal techniques, namely Cost-Benefit Analysis (CBA), Economic Internal Rate of Return (EIRR), Net Present Value (NPV), and Benefit-Cost Ratio (BCR), the project’s economic viability is assessed against its estimated completion cost of Rs. 24,948.65 crore. The evaluation finds an EIRR of 19.81% (without discounting) and 6.97% (at a 12% discount rate), both exceeding or closely approaching the discount-rate threshold; a positive NPV of Rs. 394,074.7 crore (undiscounted) and Rs. 26,136.2 crore (discounted); and a Benefit-Cost Ratio of 5.7 (undiscounted) and 1.99 (discounted), indicating that discounted benefits are nearly twice the discounted costs even under conservative assumptions. Sensitivity analysis confirms the project remains viable across tested scenarios, with EIRR never falling below 6.97% and the discounted BCR staying above 1.99. The evaluation concludes that Delhi Metro Phase 4A is economically justified, offering a cost-effective and sustainable response to Delhi’s growing urban transit demand.

Keywords: Cost-Benefit Analysis; Economic Internal Rate of Return; Net Present Value; Benefit-Cost Ratio; Metro Rail Appraisal; Delhi Metro Phase 4

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1 Project Overview

1.1 Project Description

Delhi Metro Phase 4A is a mixed underground-and-elevated metro expansion comprising one of the longest underground corridors in the Delhi Metro network. Table 1 summarises the three corridors that make up the project.

Table 1: Corridor-wise route length for Delhi Metro Phase 4A.

Corridor	Total (km)	Underground (km)	Elevated (km)
Aerocity – Tughlakabad	20.201	14.619	5.582
Janakpuri West – R.K. Ashram	28.920	7.740	21.180
Mukundpur – Maujpur	12.558	0.000	12.558
Total	61.679	22.359	39.320

Across the three corridors, the project adds 79 stations (17 underground, 62 at/above grade, Table 2) and is designed for a maximum speed of 95 km/h with a scheduled operating speed of 35 km/h.

Table 2: Corridor-wise station count.

Corridor	Length (km)	UG stations	Elevated stations	Total
Aerocity – Saket – Tughlak- abad	20.20	10	5	15
Janakpuri (West) – R.K. Ashram	28.92	7	18	25
Mukundpur – Burari – Mau- jpur	12.54	0	6	6
Total	61.66	17	29	79

1.2 Purpose

The project is intended to (i) ease traffic pressure on major roads such as the Mehrauli-Badarpur Road and NH-48 and enhance connectivity between key urban locations and the airport; (ii) serve high ridership potential, with projected daily demand reaching 19.4 lakh passengers by 2026 (1.8 million by 2031 and 2.59 million by 2041); and (iii) support sustainable urban transport by shifting travel to mass transit, thereby reducing carbon emissions, promoting energy-efficient travel, and improving air quality in Delhi.



Figure 1: Delhi Metro Rail Corporation rolling stock.

1.3 Construction Methodology

Elevated sections use viaducts of prestressed-concrete “box” girders on single piers with pile or open foundations, while the underground section is built using tunnel boring machines, with cut-and-cover construction at underground stations.

1.4 Current Status and Funding

At the time of this evaluation, 2.03 km of Phase 4A was operational, 63.07 km was under construction, 47.22 km was approved, and 10.17 km remained on hold/proposed. Funding sources for the project include the Government of India, the Delhi Development Authority (DDA), the Delhi Government, private investors, JICA, and domestic financial institutions.

2 Project Cost

The total estimated cost of the three corridors plus additional rolling stock, at January-2019 price levels including all taxes, is Rs. 22,493 crore; including Interest During Construction (IDC), the completion cost rises to Rs. 24,948.65 crore (Table 3).

Table 3: Corridor-wise estimated completion cost (Rs. crore).

Corridor	Estimated cost (Jan-2019 level, incl. taxes)	Completion cost (excl. IDC)
Aerocity – Tughlakabad	8,231	9,217
Janakpuri West – R.K. Ashram	9,349	10,357
Mukundpur – Maujpur	2,190	2,456
Additional rolling stock	2,723	2,859
Total	22,493	24,889

Table 4: Year-wise investment schedule (Rs. crore).

Year	Est. cost	constr.	Completion cost (taxes)	IDC	Completion cost (taxes+IDC)
2019–20	987.99		987.99	–	987.99
2020–21	5688.53		5954.61	4.66	5959.27
2021–22	4349.36		4730.46	9.06	4739.52
2022–23	5930.95		6710.46	18.87	6729.33
2023–24	5535.55		6505.62	26.92	6532.54
Total	22492.38		24889.14	59.51	24,948.65

3 Economic Evaluation Framework

3.1 Cost-Benefit Analysis

The economic evaluation follows standard Cost-Benefit Analysis (CBA) methodology, comparing the project’s economic costs against its economic benefits over its appraisal life.

Costs included: capital (construction) cost, land acquisition cost, and operating and maintenance costs, along with rolling-stock, environmental, utility-relocation, and contingency costs and interest during construction.

Benefits included: travel-time savings, vehicle operating cost (VOC) savings, reduced air pollution and emissions, reduction in road accidents, and fuel savings. These are summarised as annual savings in road-infrastructure maintenance costs, annual fuel savings for metro passengers and diverted road users, annual travel-time savings for metro passengers and (via decongestion) for remaining road users, annual reduction in vehicle operating costs, cost savings from reduced emissions, and reduction in accident-related costs.

3.2 Evaluation Indicators

Three principal indicators were used to assess project viability:

- **Economic Internal Rate of Return (EIRR):** the discount rate at which the present value of economic benefits equals the present value of economic costs; if EIRR exceeds the assumed discount rate, the project is economically viable.
- **Net Present Value (NPV):** the difference between the present value of benefits and the present value of costs over the project’s life; a positive NPV indicates the project adds economic value to society.
- **Benefit-Cost Ratio (BCR):** the ratio of discounted benefits to discounted costs; a BCR greater than 1 indicates benefits exceed costs.

A sensitivity analysis was additionally performed to assess how changes in traffic and cost assumptions affect project viability, using EIRR and BCR as the response indicators.

4 Economic Evaluation Results

4.1 Base-Case Indicators

Table 5 reports the base-case (2048–49 appraisal-period) economic indicator values, with and without discounting.

Table 5: Economic indicator values (base case, 2048–49).

	Without discount	With discount (12%)
Total cumulative cost (Rs. crore)	83,707.27	26,397.66
Total cumulative benefit (Rs. crore)	477,782	52,534
Benefit-Cost Ratio	5.70	1.99
Net Present Value (Rs. crore)	394,074.7	26,136.2
EIRR	19.81%	6.97%*

*A companion slide in the source deck reports 6.91% for the same discounted-EIRR figure; the discrepancy is a minor rounding/transcription difference in the original material and does not affect the conclusion that EIRR remains well above typical hurdle rates for public transit investment.

Since EIRR (19.81% undiscounted) exceeds the 12% discount rate used, and the NPV is positive under both undiscounted and discounted assumptions, the project is economically viable: for every Rs. 1 invested, the undiscounted analysis projects Rs. 5.7 of benefits, and even under the more conservative 12%-discounted analysis, benefits are projected at nearly double the costs (BCR = 1.99).

A separate Financial Internal Rate of Return (FIRR) was also estimated: 8.03% without Transit-Oriented Development (TOD) and Value Capture Financing (VCF) income, rising to 9.60% when TOD/VCF revenue streams are included, indicating that non-fare revenue mechanisms materially improve the project’s financial (as distinct from economic) return.

4.2 Sensitivity Analysis

Sensitivity analysis assessed how changes in traffic and cost assumptions affect project viability. The project remained viable across the scenarios tested, with EIRR never falling below 6.97% and the discounted Benefit-Cost Ratio never falling below 1.99, indicating that the economic case for Phase 4A is robust to reasonable variation in underlying assumptions.

5 Project Benefits

Project benefits are classified as direct and indirect (Table 6).

Table 6: Direct and indirect project benefits.

Direct benefits	Indirect benefits
Expedited commutes	Job creation
Reduced congestion	Business growth
Vehicle cost savings	Greener environment
Economical transport	Property appreciation
Enhanced safety	Improved air quality
Fare revenue	Economic development

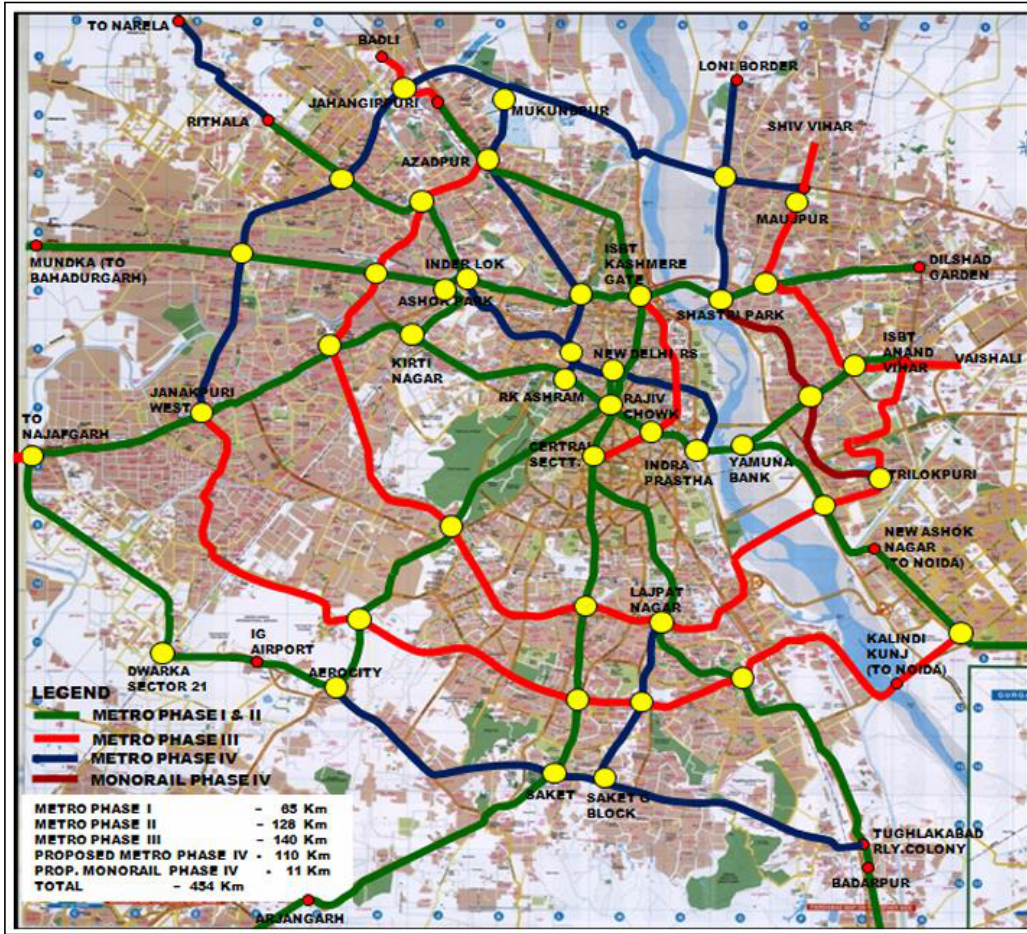


Figure 2: Delhi Metro network, showing Phase I-IV and proposed monorail corridors.

6 Conclusion and Recommendations

6.1 Economic Viability Summary

- **EIRR (19.81%)** exceeds the 12% discount-rate threshold, indicating strong socio-economic benefits such as reduced travel time, fewer accidents, and environmental gains.
- **NPV is positive** under both discounted and undiscounted assumptions, reinforcing the project's long-term economic viability.

- **Benefit-Cost Ratio** of 5.71 (undiscounted) and 1.99 (at 12% discount) shows that even under conservative assumptions, benefits substantially exceed costs.
- **Sensitivity analysis** confirms the project remains viable across scenarios tested, with EIRR never falling below 6.97% and discounted BCR staying above 1.99.

6.2 Key Takeaways for Decision-Makers

A strong EIRR and BCR confirm the economic viability of the project; earlier completion yields higher cumulative benefits (as delay defers benefit realisation while cost escalation continues); the project is critical for sustainable urban growth in the National Capital Region; the corridor alignment strategically links major employment, residential, and transit nodes; and there is an opportunity for policy synergies through last-mile connectivity investment alongside the metro expansion.

6.3 Learnings

This exercise provided hands-on experience in economic evaluation techniques, namely Cost-Benefit Analysis, EIRR, and NPV, and in interpreting sensitivity analysis to assess project risk under uncertainty. It also highlighted the real-world impact of large transit infrastructure investment on urban mobility and the importance of data-driven planning in public transport project appraisal.

6.4 Concluding Remarks

Delhi Metro Phase 4A is economically justified, not only meeting but significantly surpassing standard evaluation thresholds. It stands as a cost-effective, sustainable response to Delhi's growing urban transit demand, illustrating how thoughtful transport planning can deliver economic, environmental, and social benefits simultaneously, while underscoring the need for inter-agency coordination and timely execution to realise its full potential.

References

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